

Skills

Programming languages: C++ (Qt), C#, C, Python (PyTorch), Kotlin, TypeScript, JavaScript (Node & React & Next), HTML, CSS, Java
Tools: Unreal Engine, Unity, MATLAB, Android Studio, Git, Jira, Google Test, Figma, LaTeX, Linux, Windows

Education

University of Waterloo - Candidate for Bachelor of **Computer Science (AI Specialization)**

2021 - 2026

- Cumulative GPA: **87% + Term Distinction Honour** x 6
UW CS International Student Upper-Year Scholarship + UW President's Scholarship

Experience

Christie Digital Systems - Application Software Developer

Sep - Dec 2024

- Developed a **black level uniformity correction algorithm** with **MATLAB** to address brightness and color inconsistencies in laser projectors; Integrated into Hawkeye calibration software using **C++** and **Qt framework**, improving the color accuracy by over **30%**.
- Enhanced Hawkeye's **web UI (QT/JavaScript)** by adding dynamic control sliders for **real-time** RGB laser drive level adjustments.

Ford Motor Company - Embedded Software Developer

Jan - Apr 2024

- Maintained the **FNV4 Data Collection Platform's C++** codebase, improved integration with **Autosar Adaptive** and **SOA-based** systems while optimizing **CCS privacy controls** to ensure regulatory compliance, reduced data scrubbing latency by **15%**.
- Developed a **multi-threaded** data ingestion pipeline with **JavaScript** sandboxing to handle safe execution of **over 1,000** concurrent intents, optimizing resource management which reduced vehicle sensor data processing latency by **23%**.

Behaviour Interactive Inc. - Software Engineer (Dead by Daylight Project)

May - Aug 2023

- Developed **Unreal Engine** components using **C++** for **real-time**, in-editor actor **collision detection** with customizable visual indicators, streamlined level design and boosted team productivity by **~40%**.
- Optimized a custom **Unreal Engine plugin (Tile Editor)** to automate the conversion of over **700** game scene tiles from blueprints to levels, which saved over **100** hours of manual work and boosting design team efficiency by **~30%**.

University of Waterloo - Instructional Support Assistant

Aug - Dec 2022

- Performed **UI** and **API quality assurance tests** for the **Android** SES application, identifying and reporting bugs for timely resolution.
- Constructed **C++** configuration scripts for the auto-grading system (**Lint R**) to automated assignment assessments and feedback.
- Conducted weekly office hours, tutorials, and pre-exam review sessions. (Praised by students as "**encouraging** and **supportive**".)

Projects

UW – Research Assistant (URA) (HCI – Around Ear Gestures for VR Locomotion)

2024

- Collaborated with Prof. **Jian Zhao** to design and implement a **VR** game in **Unity** with mazes and interactable objects to test **around-ear bare-hand gestures** for VR locomotion, improving **accessibility** for seated and lower-limb disabled users.
- Programmed VR features in **C#**, including **locomotion mechanics** (movement, jumping, crouching), **humanoid model** animations synced with player actions, and **interactions** with game objects, enhancing the immersive gameplay experience.

Collaborative Whiteboard Android App (YADA)

2023

- Used **Kotlin** and **Jetpack Compose** in **Android Studio** to design and build a responsive **Android whiteboard App** in a team of 4, supporting **real-time**, **low-latency** collaboration, allowing multiple users to draw and interact on the whiteboard simultaneously.
- Implemented dynamic drawing tools (**pen**, **eraser**, and **shape** features), with **undo/redo** functionality for a seamless user experience.

The Game of ChamberCrawler3000+ (CC3K+)

2023

- Built the game in **C++** with **OOP**, leveraged **smart pointers** and **STL** for efficient memory management and dynamic gameplay logic.
- Enhanced gameplay interaction by implementing **Observer** and **Decorator Design Patterns**, added key features such as NPC combat, merchant trading, and prop equipping.
- Elevated the gaming and visual experience by implementing a series of "cheat codes" via **C++ macros** and a **colored text display UI**.

WLP4 (Simplified C Language) Compiler Project

2023

- Programmed a full-featured compiler for WLP4 using **C++** and a **simplified Maximal Munch algorithm** for **lexical analysis**.
- Implemented advanced features such as **nested procedures** and **closures** while conducting **context-sensitive semantic analysis**.
- Enhanced runtime performance via **in-line procedure substitution** and **constant folder** and **propagation** optimizations.

UW Course Scheduler (UWCS)

2023

- Developed a smart scheduling agent for UW students using **Python**, leveraging **Selenium** and **Requests** to extract class data and **BeautifulSoup** for analysis; Adopted by **200+** users with **500+** successfully generated schedules.
- Combined a conflict-free scheduling algorithm with **OpenAI GPT-3.5 API** integration for personalized schedule generation, supporting **real-time editing** and **GUI calendar display**, which significantly improved user experience and simplified course selection process.